

electron/workspace-history.cjs

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Overview

`electron/workspace-history.cjs` adds snapshotting and indexing for workspace commit history. The file now imports Node's `fs/promises`, `fs`, `path`, and a local `runGit` helper (R3–R6).

It also pulls `upsertWorkspace` and `getWorkspace` from `workspace-store.cjs` (R33). New helper functions are defined: `removeDirSafe` (R8–R16), `isUsableSnapshotDir` (R18–R32), `listCommitsOldestFirst` (R35–R58), `snapshotCommit` (R61–R78), and `ensureCommitSnapshot` (R80–R98).

The main workflow, `indexWorkspaceHistory`, iterates over all commits, creates snapshots, writes metadata, and updates the workspace's history status (R100–R183). All public functions are exported (R185–R191).

Key changes

- **Imports:** `fsp`, `fssync`, `path`, `runGit` (R3–R6).
- **Workspace store hooks:** `upsertWorkspace`, `getWorkspace` (R33).
- **Utility functions:** `removeDirSafe` (R8–R16), `isUsableSnapshotDir` (R18–R32).
- **Git helpers:** `listCommitsOldestFirst` (R35–R58), `snapshotCommit` (R61–R78), `ensureCommitSnapshot` (R80–R98).
- **Indexing workflow:** `indexWorkspaceHistory` (R100–R183).
- **Exports:** all functions (R185–R191).

Impact

- **Correctness:** `snapshotCommit` creates a detached worktree per commit, isolating each snapshot.
- **Maintainability:** Snapshot logic is centralized, easing future extensions.
- **Performance:** Worktree creation per commit can be costly; progress is reported via `sendProgress`.
- **Compatibility:** Uses Node's `fs/promises` and `fs` APIs, suitable for all Electron targets.
- **Observability:** Workspace status fields and progress callbacks provide UI hooks.

Risks & follow-ups

- **Git command failures:** `runGit` errors are not caught in `snapshotCommit`; ensure failures surface to the UI.
- **Race conditions:** Concurrent `indexWorkspaceHistory` calls could corrupt `workspace.historyStatus`; consider idempotency or locking.
- **Path resolution on Windows:** `path.resolve` may produce backslashes; verify downstream consumers handle them.

- **Test coverage:** No unit tests exist for this module; add mocks for `runGit` and the file system to validate edge cases (e.g., missing repo, empty snapshot dir).